

Continuous boundary data do not rescue interval square roots in $C(K)$

Candidate counterexample packet for arXiv:1604.01211

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Status: candidate full counterexample, likely valid, subject to human review.
For every compact Hausdorff space containing a nontrivial convergent sequence, there are continuous f, g satisfying the source paper's necessary pointwise conditions $g \geq 0$ and $|f| \leq g$, but $[f, g]$ has no square root which is an interval. Consequently, among first-countable compacta, finiteness is exactly the condition under which those necessary conditions suffice for all continuous boundary data.

1 The source question and the scoped answer

For a compact Hausdorff space K , an interval in the real Banach algebra $C(K)$ is

$$[f, g] = \{h \in C(K) : f \leq h \leq g\}.$$

Moreno and Schneider call an interval I a square root of $[f, g]$ when $I \cdot I = [f, g]$, with pointwise multiplication. In Section 5 they ask:

“Which intervals have a square root that is an interval?”

They prove that $g \geq 0$ and $|f| \leq g$ are necessary, that $f \geq 0$ is sufficient, and that $-g \leq f \leq 0 < g$ is sufficient when g is continuous [1, Proposition 5.1]. The result below shows that the necessary conditions are not sufficient even when *both* boundary functions are continuous. It also gives a complete answer to the corresponding uniform sufficiency question for first-countable compacta.

Figure 1 reproduces the source question and the neighboring necessary and sufficient conditions from printed page 14.

–1. Since h is continuous, this is a contradiction. It shows that the space K is finite. $\square$

5. SQUARE ROOTS OF INTERVALS

A *convex body* in $C(K)$ is a nonempty bounded closed convex subset of $C(K)$. In [11], Theorem 4.3, the following was shown. If $C, D \subset C(K)$ are convex bodies such that the closure of the sum $C+D$ is an interval, then each of C and D is an interval. One may ask whether an analogous result is true for products instead of sums. But it is easily seen that this is not the case (see the example below).

Now we restrict ourselves to products of a convex body with itself, and we say that the convex body C is a *square root* of the interval $[f, g]$ if $C \cdot C = [f, g]$. It is easy to see that an interval can have a square root that is not an interval, provided that K is not a singleton. For example, let x_0, x_1 be distinct points of K , let

$$A = \{f \in [-\bar{1}, \bar{1}] : f(x_0) \geq 0\}, \quad B = [-\bar{1}, -\bar{1}].$$

Then A and B are convex bodies in $C(K)$. Let C be the convex hull of $A \cup B$; it is closed. Then $C \cdot C \subset [-\bar{1}, \bar{1}]$ trivially. Conversely, $f \in [-\bar{1}, \bar{1}]$ satisfies either $f(x_0) \geq 0$ and hence $f = f \cdot \bar{1} \in C \cdot C$, or $f(x_0) < 0$ and hence $f = (-f) \cdot (-\bar{1}) \in C \cdot C$. Thus, C is a square root of $[-\bar{1}, \bar{1}]$. On the other hand, C is not an interval. If it were an interval, it could only be the interval $[-\bar{1}, \bar{1}]$, but a function $h \in [-\bar{1}, \bar{1}]$ with $h(x_0) = -1$ and $h(x_1) = 1$ (which exists by Urysohn's lemma) does not belong to C .

This raises the question: which intervals have a square root that is an interval. Trivially, this holds for intervals $[f, g]$ with $f \geq 0$. In fact, if $f \geq 0$ and if $H \in [f, g]$, then $\sqrt{H} \in [\sqrt{f}, \sqrt{g}]$, which shows that $[\sqrt{f}, \sqrt{g}] \cdot [\sqrt{f}, \sqrt{g}] = [f, g]$.

The following result provides additional information, which will be used below to show that also an assumption on the existence of square roots leads to finite spaces.

Proposition 5.1. *Let $[f, g]$ be an interval in canonical representation in $C(K)$. In order that $[f, g]$ have a square root which is an interval, it is necessary that*

$$(5.1) \quad g \geq 0 \quad \text{and} \quad |f| \leq g,$$

and it is sufficient that g is continuous and

$$(5.2) \quad -g \leq f \leq 0 < g.$$

Figure 1: The square-root question and Proposition 5.1 in the source paper.

2 Idea of the proof

Take $g \equiv 1$ and make a continuous lower boundary f alternate between small positive and small negative values along a sequence converging to x_0 . If an interval I squared to $[f, 1]$, then the member f itself would factor as $f = hk$ with $h, k \in I$.

At a point where $f = a > 0$, the scalar interval $I(x)$ cannot cross zero, and attaining the lower endpoint a forces both factors to have modulus $\sqrt{a}$. Hence both h and k tend to zero along the positive subsequence. At a point where $f = a < 0$, attaining the lower endpoint of the scalar square forces one factor to be an endpoint of modulus 1. Along the negative subsequence, therefore, one of h, k always has modulus 1. These two conclusions are incompatible with continuity at x_0 .

3 The scalar endpoint obstruction

Lemma 1. *Let $-1 \leq a \leq 1$, $a \neq 0$, and let $J = [c, d]$ be a nonempty compact real interval such that*

$$J \cdot J = [a, 1].$$

If $u, v \in J$ and $uv = a$, then:

1. *if $a > 0$, one has $|u| = |v| = \sqrt{a}$;*
2. *if $a < 0$, one has $\max\{|u|, |v|\} = 1$.*

Proof. Suppose first that $a > 0$. Since $J \cdot J$ contains neither zero nor a negative number, J lies strictly on one side of zero. If $0 < c \leq d$, then $J \cdot J = [c^2, d^2]$, so $c^2 = a$, $d^2 = 1$, and $uv = a$ forces $u = v = c$. If $c \leq d < 0$, then $J \cdot J = [d^2, c^2]$, and the symmetric argument gives $u = v = d$. This proves the first assertion.

If $a < 0$, then $c < 0 < d$ and the lower endpoint of $J \cdot J$ is $cd = a$, whereas its upper endpoint is $\max\{c^2, d^2\} = 1$. For $uv = a$, the two factors have opposite signs. Assuming $u < 0 < v$, we have $|u| \leq |c|$ and $v \leq d$; equality in $|u|v = |a| = |c|d$ forces $u = c$ and $v = d$. The other ordering is identical. Thus $\max\{|u|, |v|\} = \max\{|c|, d\} = 1$. $\square$

4 Continuous counterexamples

Theorem 1. *Let K be a compact Hausdorff space containing a sequence of distinct points (x_n) converging to a point $x_0 \notin \{x_n : n \geq 1\}$. There exists a continuous function $f : K \rightarrow [-1, 1]$ such that*

$$1 \geq |f|,$$

but the continuous interval $[f, 1]$ has no square root which is an interval in $C(K)$.

Proof. The set $S = \{x_0\} \cup \{x_n : n \geq 1\}$ is closed in K . Define

$$r(x_0) = 0, \quad r(x_n) = \frac{(-1)^n}{n+1}.$$

This is a continuous function $S \rightarrow [-1, 1]$. Compact Hausdorff spaces are normal, so the Tietze extension theorem gives an $f \in C(K, [-1, 1])$ extending r [2, Section 33].

Assume for contradiction that an interval I satisfies

$$I \cdot I = [f, 1].$$

Replace any presentation of I by its canonical presentation. By [1, Lemma 3.1(i)], evaluation at each $x \in K$ gives a scalar interval $I(x)$ satisfying

$$I(x) \cdot I(x) = [f(x), 1].$$

Since $f \in [f, 1] = I \cdot I$, there are $h, k \in I$ with $hk = f$.

For even n , $f(x_n) > 0$. Lemma 1 gives

$$|h(x_n)| = |k(x_n)| = \sqrt{f(x_n)} \rightarrow 0.$$

Continuity along the even subsequence implies $h(x_0) = k(x_0) = 0$. For odd n , however, $f(x_n) < 0$, and the same lemma gives

$$\max\{|h(x_n)|, |k(x_n)|\} = 1.$$

Continuity of h and k at x_0 makes the left-hand side tend to zero along the odd subsequence, a contradiction. $\square$

The explicit metrizable example is $K = [-1, 1]$, $f(t) = t$, and $g(t) = 1$. Thus no extension theorem is needed for the simplest instance.

Corollary 1. *For a first-countable compact Hausdorff space K , the following are equivalent:*

1. K is finite;
2. every continuous interval $[f, g]$ satisfying $g \geq 0$ and $|f| \leq g$ has a square root which is an interval.

Proof. If K is infinite, compactness supplies an accumulation point. A countable local base at that point lets one choose a sequence of distinct points converging to it, so Theorem 1 disproves (2).

If K is finite, $C(K)$ is a finite Cartesian product of copies of $\mathbb{R}$. For each $x \in K$, the scalar condition $g(x) \geq 0$, $|f(x)| \leq g(x)$ produces a scalar interval whose square is $[f(x), g(x)]$ (the elementary classification recorded in the proof of [1, Proposition 5.1]). Choosing one such interval independently at each of the finitely many points gives an interval in $C(K)$ whose square is $[f, g]$. $\square$

5 Scope, novelty check, and review recommendation

The theorem does not classify individual square-rootable intervals on an arbitrary compactum. It decisively rules out the most direct completion of the source proposition—sufficiency of its pointwise necessary inequalities, even for continuous boundary data—and Corollary 1 fully characterizes the first-countable compacta on which that uniform sufficiency statement holds.

On 11 August 2026, exact arXiv-id/title searches in the run registry and all cheap run indexes found no prior packet. Searches of the locally parsed arXiv corpus for the source title, “square roots of intervals,” and the necessary conditions found only the source. Bounded web searches for the exact question, the source title with “square root,” and close $C(K)$ variants likewise found the 2016 paper but no later resolution or this continuous-boundary obstruction. This is a bounded novelty check, not an exhaustive priority claim.

Human review should focus on the use of source Lemma 3.1(i) to identify every pointwise scalar square and on the equality cases in Lemma 1. The explicit $[-1, 1]$ example isolates the entire analytic obstruction and can be checked independently of the Tietze-extension strengthening.

References

- [1] J. P. Moreno and R. Schneider, *Multiplication of convex sets in $C(K)$ spaces*, Studia Math. 232 (2016), 173–187; arXiv:1604.01211.
- [2] J. R. Munkres, *Topology*, 2nd ed., Prentice Hall, 2000.